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Introduction

This project uses computational text analysis methods to investigate how the concept of "trade" is used across a corpus of 571 Early Modern documents. The word "trade" is particularly interesting to study because it is polysemous and can refer to a person's individual occupation or craft, or to large-scale commercial activity between nations and institutions. To explore the meaning of “trade” in my corpus, I use a combination of four methods: keyword extraction to locate all occurrences of the term, Named Entity Recognition (NER) to identify the people, places, and institutions most associated with it, BERT-based clustering to uncover contextual variation in how the word is used, and logistic regression to build a classifier that can distinguish between its two primary meanings. Together, these methods allow me to understand how “trade” is used semantically across Early Modern English texts.

Literature Review

The use of computational methods to analyze meaning in text has generated a substantial body of relevant literature. I will draw on work relating to word embeddings, contextual language models and supervised classification to form the theoretical basis of my project.

Foundational work by Mikolov et al. (2013) established that words could be encoded as dense vectors in a continuous space, where proximity reflects semantic and syntactic similarity. Their skip-gram model demonstrated that meaning could be operationalized distributionally: a word's significance derives from the company it keeps across a large corpus. However, the Word2Vec family of models assigns each word a single fixed representation regardless of context, a limitation later identified by subsequent researchers that can have a large consequence when analyzing lexically rich texts. For example, this project focuses on analyzing the concept "trade" in Early Modern literature and trade can hold multiple meanings (polysemy) ranging from a person's livelihood to a nation's commercial policy within the same document. These differences in meaning may not be revealed without factoring in context.

Peters et al. (2018) addressed this shortcoming directly by introducing ELMo, a deep bidirectional language model that generates distinct representations for the same word depending on its surrounding sentence. The paper showed how higher model layers capture semantic information while lower layers encode syntactic patterns and illustrated this using the word "play," which is polysemous and can receive meaningfully different vector representations across different contexts. This capacity to model polysemy provides the theoretical foundation for the BERT-based analysis which I use to accomplish step 3 of this project.

Building on Peters et al., Rogers, Kovaleva & Rumshisky (2020) synthesized over 150 studies of BERT's internal behavior, providing a comprehensive account of what contextualized models do and do not learn. Their research confirmed that BERT's embeddings form distinct clusters corresponding to different word senses, making the model well-suited for studies of semantic variation. However, they also document BERT's limitations, showcasing that BERT's semantic knowledge is unevenly distributed across layers, meaning it struggles with numbers and named entity generalization. Further, its performance degrades on

text that differs substantially from its pretraining data. Since BERT was primarily trained on contemporary English text sources, this could have impacts on the results of this project which draws from a corpus of Early Modern documents that contain vocabulary and follow spelling conventions as well as syntactic structure that is quite different from modern English.

Lastly, Barberá et al. (2021) provide a methodological guide to supervised machine learning for text analysis that informs this project's approach to supervised classification. The paper's key finding is that machine learning models consistently outperform simpler dictionary-based methods at categorizing text, as long as the analyst carefully builds a large enough set of labeled training examples. They also show that it is better to label as many different documents as possible rather than labeling the same documents multiple times. Finally, the paper cautions that decisions made early in the process, such as how documents are selected and how the search is set up, can significantly affect the final results. This is especially relevant here, given that 17th-century texts use archaic spellings and vocabulary that can make even a simple keyword search more complicated than it would be with modern text.

Together, these four sources trace a clear path from the basics of word representation, through contextual meaning, to practical text classification and are directly applicable to this project and its methods.

Step 1: Concept and Data Preparation

Concept Selection and Rationale

The concept selected for this analysis is trade, a term that appears frequently in Early Modern economic and travel writing and can have a range of potential meanings from the physical exchange of goods between merchants and ports, to broader discussions of commercial policy, national wealth, and economic relationships between states and peoples. Because the word "trade" and its variants (trades, traded, trading, trader, traders) all point to the same underlying concept, all forms were included in the search using a case-insensitive regular expression designed to capture all grammatical variants while avoiding partial matches such as "tradesman."

Corpus and extraction

Across the 571 documents in the corpus, 9401 occurrences of "trade" were identified. The word “trade” was also mentioned in only 380 of the 571 documents. Further, the unit of analysis I chose for the project was the sentence that contained each keyword occurrence, providing enough context to understand usage without over-expanding the analytical window.

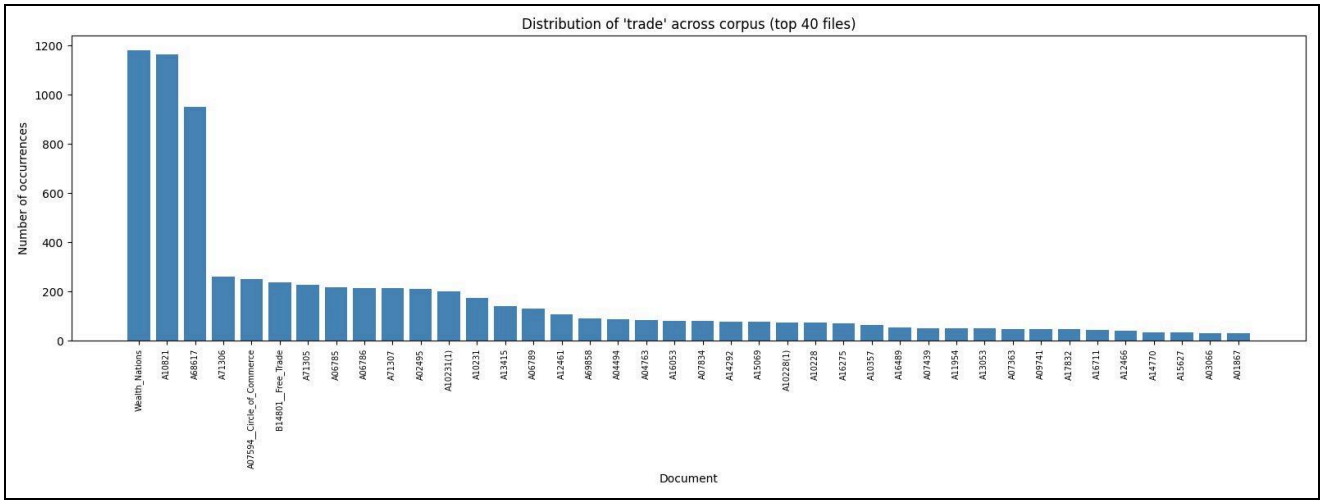

Figure 1. Documents with the top 40 occurrences of "trade" in the corpus.

Initial Patterns

After conducting step 1 of the analysis, I noticed that the distribution of “trade” is notably uneven across the texts. Over 3,300 occurrences of “trade” are concentrated within just three documents: Wealth of Nations (1185 occurrences), A10821 (1165 occurrences), and A68617 (952 occurrences). This could suggest that a small number of texts are particularly concerned with looking at trade as a subject while a larger number of the remaining documents use the term “trade” incidentally rather than as a primary discussion point.

Step 2: Named Entity Recognition (spaCy)

Method and Focus

I ran spaCy's en_core_web_sm model on all 9,401 sentences with the word “trade” mentioned to investigate how the corpus discusses it. For each sentence, the model identified and labeled all named entities. Trade is dependent on institutional structures that dictate how commercial transactions occur and on collective actors who both guide and carry out the process. Further, it is fundamentally geographic in many contexts and involves the transfer of goods from one area to another. Lastly, a third important facet of trade is the people who actually engage in it. For this reason I chose to center my analysis primarily on GPE/LOC (or places), but also look into ORG (institutions) and PERSON since they delivered a higher frequency count.

Entity Frequency Table

Entity Type	Count
ORG	8,777
PERSON	7,153
GPE	6,770

CARDINAL	4,476
NORP	3,019
DATE	1,750
LOC	1,343
ORDINAL	1,100
PRODUCT	622
QUANTITY	466
LANGUAGE	403
WORK_OF_ART	356
TIME	275
FAC	249
MONEY	80
LAW	60
EVENT	39

Co-occurrence between entities and "trade"

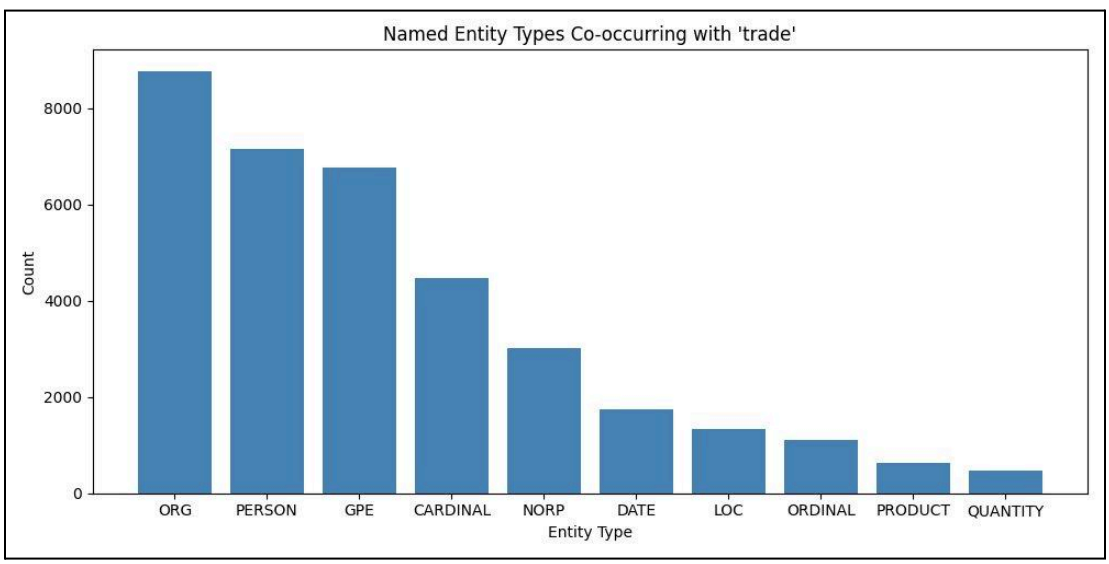

Figure 2. Bar chart of named entity type frequencies co-occurring with "trade."

Is the concept associated with specific people, institutions, or places?

"Trade" in this corpus definitely seems to be associated with specific places. Co-occurrence between geographic distribution (GPE/LOC) and "trade" revealed several names of places, the most prevalent of those being England (417), Portugal (307), India (253), France (164), Sea (159), London (130), Europe

(130), Persia (128), Kingdome (115), China (110), Great Britain (95), Russia (83), Holland (75), Moha (73), and Venice (57). These specific names give us insight into where, and between what places, was trade occurring at the time these documents were written and even where the authors were situated. Further, the inclusion of the terms “Sea” and “Kingdome” also reveal that the trade could be maritime in nature or that people still referred to empires (for example, the British empire) as kingdoms. Lastly, the terminology used for the names of the places themselves is indicative of the time period the documents were written in. (for example, the use of “Persia” which is now called Iran).

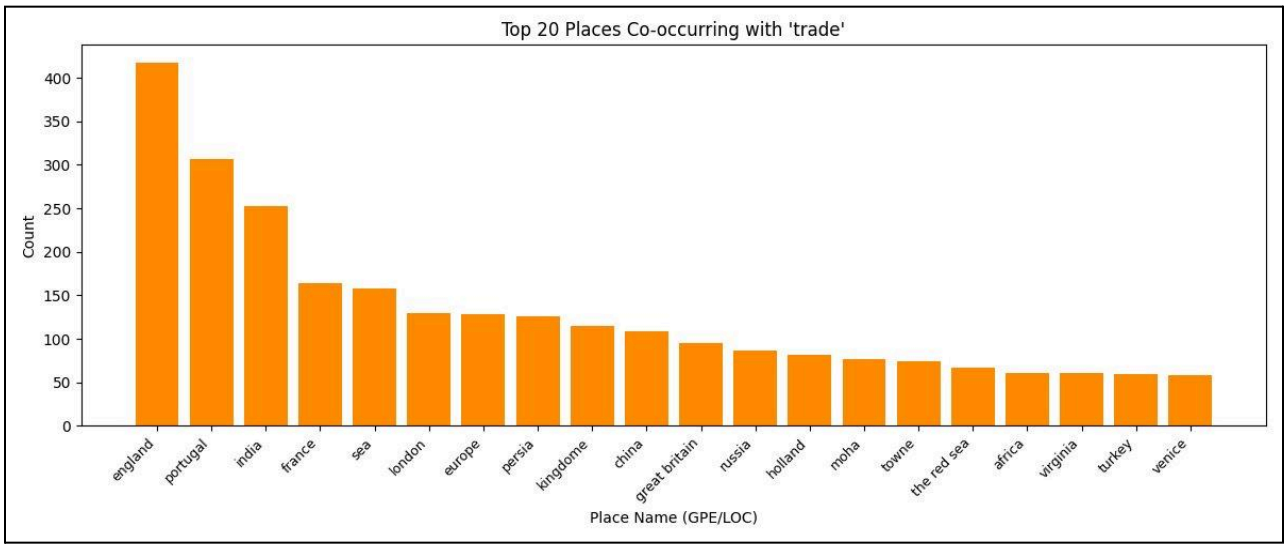

Figure 3: co-occurrence between GPE/LOC and “trade”

Do different entities correspond to different uses of the concept?

When I pulled random examples for each entity type, GPE sentences tended to describe trade as something happening at a specific place, for example, a ship arriving, a merchant negotiating, a journey being made. On the other hand, ORG sentences tended to frame trade more abstractly, as a social condition or matter of policy. Further, PERSON sentences were more mixed. This suggests that the entity type co-occurring with "trade" does correspond, at least loosely, to different registers of usage.

<u>Examples sentences that demonstrate different uses of trade:</u>

GPE: Entity: england | File: A68617.txt

Sentence: “the mogul inconstancy and captain Hawkins departure with Sr Henry MIDDLETON to the red-sea thence to Bantam and after for England all my go and send to Mocrebchan for my money or cloth be in vain I be abuse so base by he that I be force to demand justice of the king who command that the money be bring before he but for all the king command he do as he list and do what I can he cut i off twelve”

- ➔ This shows trade as a journey

PERSON: Entity: frenchmen | File: A15072.txt

Sentence: “Whereby it may be well conceiued, that the fishing which the Frenchmen haue there, is a very profitable trade; seeing they not onely saile thither in the hardest time of Winter, which is aboue 650”.

- ➔ This shows trade as an occupation

Limitations of NER in this corpus

The results reveal several limitations. First, the model performed poorly on categorising or tagging. Words like knowne, corne and hee are incorrectly tagged as GPE/LOC entities, which also distorts the GPE/LOC frequency counts. Further, "trade" itself appears 837 times as an ORG entity which showcases that the model frequently misclassifies the concept itself as an organization. Second, the spaCy model used for this project is quite small, and a larger model that was specifically trained on early modern English would be able to produce more accurate results.

Step 3: Contextual Meaning with BERT

Method

I used BERT to generate contextual embeddings to analyze how the meaning of "trade" varies across the corpus. I chose to analyze a random sample of 1,000 sentences, which was drawn from the full set of 9,401 occurrences, for computational feasibility. For each sentence, the embedding corresponding to the "trade" token in BERT's final hidden layer was extracted. When the tokenizer did not produce an exact match, the [CLS] or classification token embedding was used as a fallback. All embeddings were L2-normalized before clustering. KMeans clustering (k=2) was then applied to group the embeddings into two contextual clusters. To visualize the results, Principal Component Analysis (PCA) reduced the 768-dimensional embeddings to two dimensions. The choice of two clusters was motivated by the expectation (informed by Step 2's NER results) that trade operates across at least two broad registers (occupation and commercial activity/journey) and that further subdivision within those registers was likely.

PCA Visualization and Cluster

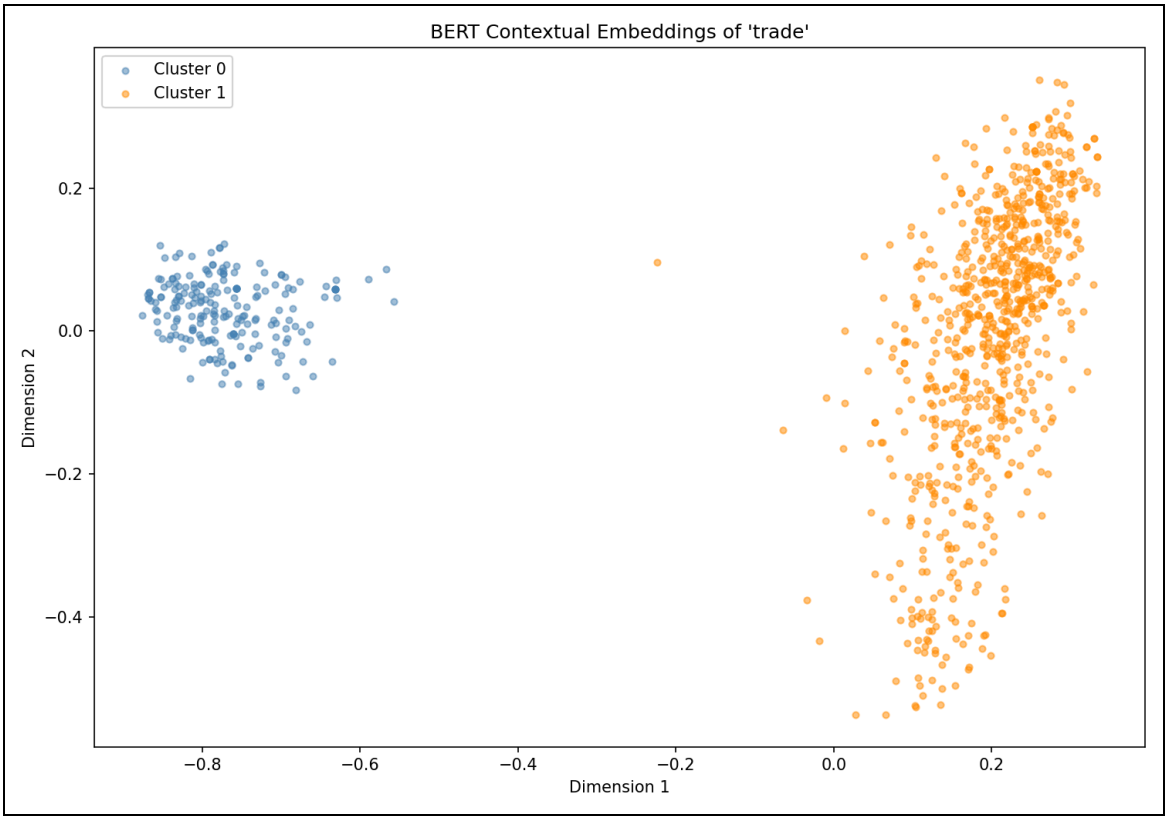

Figure 4. PCA visualization of BERT contextual embeddings of "trade" across 1,000 sampled sentences, colored by KMeans cluster (k=2)

The PCA plot shows a very clear separation between the two clusters. Cluster 0 (blue, 206 sentences) sits entirely on the left side of the plot, forming a tight compact group. Cluster 1 (orange, 794 sentences) occupies the right side and is much more spread out vertically. The large gap between the two clusters along the horizontal axis suggests that BERT identifies a consistent difference between cluster 1 and 2.

Do you observe distinct meanings?

I do observe a distinct meaning between the two clusters after reading through random examples from each cluster. The two clusters correspond largely to different uses of "trade:" Cluster 0 tends to contain sentences where "trade" is used as a specific personal occupation or concrete activity taking place. For example:

- *"Goe but to a Printer, and you shall heare many strange words of his inuention and misterie" (A13118)*
- *"All men of all kindes of trades for the most part, sell honesty, trueth, consc[ience]" (A04845)*
- *"Their chiefe townes for fish are, Yaslaue, Bealozera... which all yeeld a large custome to the Emperour euery yeere for their trades of fishing" (A02495)*

Cluster 1 is larger and contains sentences where "trade" tends to be used in a more institutional or policy-oriented sense. For example:

- *"This gain is procured by Merchants, by or under the three essential parts of Traffic, either by Commodities, or by Moneys, or by exchange of moneys, according to every man's profession and occasion in the course of Trade" (A06785)*
- *"It discourages the exportation of the materials of manufacture, and of the instruments of trade, in order to give our own workmen an advantage, and to enable them to undersell those of other nations in all foreign markets" (Wealth_Nations)*
- *"When the clothiers shall haue this commodiousnesse, they will increase their trade, and set more people on worke to make good clothes... to the generall aduancement of trafficke, his maiesties Customes" (A06786)*

How do you interpret the differences between contexts?

Cluster 0 tends to use "trade" to describe what a specific person or group does — a printer's trade, a fisherman's trade, a craftsman's trade whereas Cluster 1 tends to use "trade" to describe something happening at a larger scale, involving merchants, nations, commodities, and policy. This matches with what I identified during Step 2.

Further, Cluster 1 is much larger and vertically spread out than cluster 0, which is smaller and compact. This suggests that while k=2 revealed the most distinct uses of trade, a higher k would potentially reveal more variation within cluster 1 since the vertical spread and size indicate that it potentially contains greater semantic variation.

Step 4: Supervised Classification

How well does the model distinguish meanings?
The model achieved 95% accuracy, however, the confusion matrix gives us more insight into this. The model did very well for the Policy/Commerce sentences, correctly classifying all 805 sentences without confusing them for an occupation sentence, However, for the Occupation sentences, the model only classified 33 out of 77 sentences correctly and misclassified the remaining 44 as policy/commerce sentences.

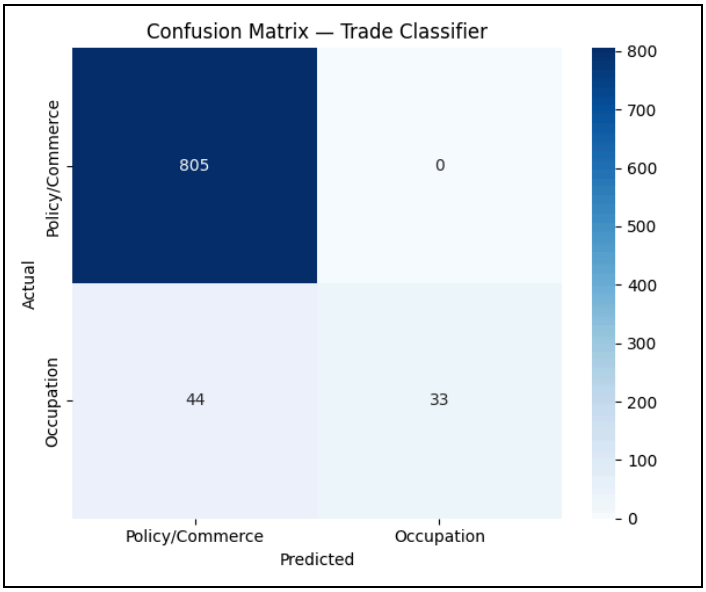

Figure 5: confusion matrix

What features seem to drive classification?

The features that push sentences toward Policy/Commerce are words like port, merchant, profit, nation and commerce, aligning with the broader idea of these sentences talking about "trade" as a policy topic or as a commercial venture. Another striking feature that aligns with this idea is "foreign trade". On the other hand, features pushing sentences toward Occupation are words like occupation, profession, calling, labor, vocation, smith, carpenter, and weaver.

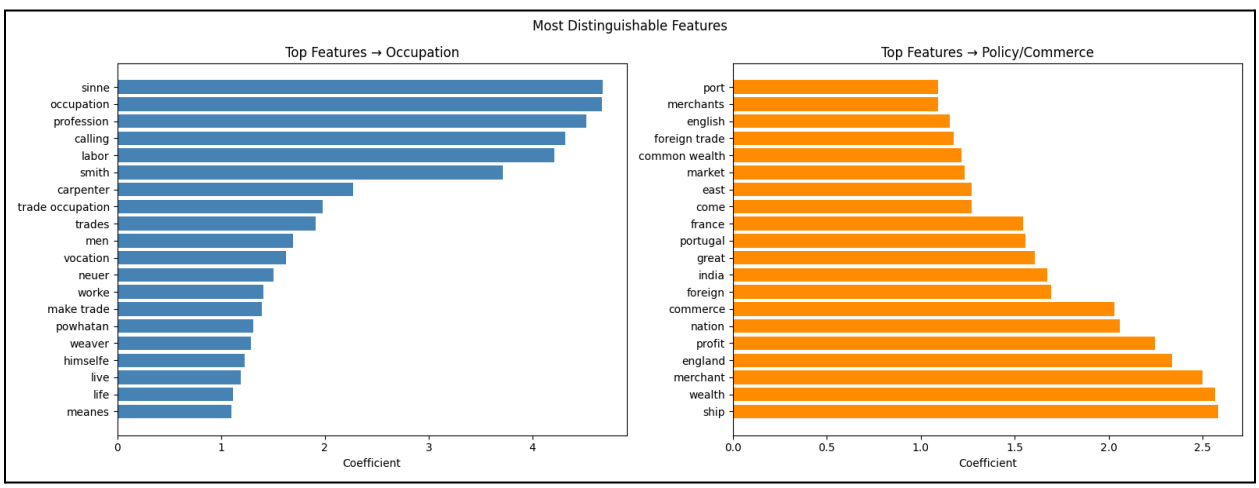

Figure 6: representative examples for Policy/Commerce and Occupation sentences

Where does the model fail?

The model's failure is the 44 Occupation sentences that get misclassified as Policy/Commerce sentences. These are probably sentences where the occupation-related signal is weak or where the sentence happens to also contain a word that appears in the Policy/Commerce vocabulary. The model is very conservative

about labelling something as Occupation/Craft and only does so when the signal is very clear, which is why precision for that class is 1.00 but recall is only 0.43.

Step 5: Synthesis

Through this project, I aimed to find out what the word “trade” meant lexically in the corpus of Early Modern texts. The clearest finding is that "trade" primarily relates to policy or commercial activities in the corpus. When the model was applied to all 9,401 sentences, 9,185 were classified as Policy/Commerce and only 216 as Occupation. This is consistent with the NER results, which showed that the places most frequently co-occurring with "trade" are major nations and trading regions like England, Portugal, India, France, and China, which are the geographic vocabulary of international commerce rather than individual skilled work. It is also consistent with the BERT clustering in Step 3, where Cluster 1, the larger and more spread out group, contained sentences oriented around merchants, commodities, national wealth, and policy, while Cluster 0, the smaller and more compact group, contained sentences where trade referred to a specific personal activity or work.

The usage of trade as an ‘Occupation’, while a minority, is still distinct within the corpus. The logistic regression model identified a distinguishable vocabulary for this with words like occupation, profession, calling, labor, vocation, smith, carpenter and weaver that do not apply to Policy/Commerce. They also differ greatly from the Policy/Commerce vocabulary that contained words like ship, wealth, merchant, england, profit and nation. The fact that the classifier achieved perfect precision (1.00) for the Occupation category means that when it did predict this label, it was never wrong. However, the problem here was with recall since the model missed 44 out of 77 sentences.

Taken together, these findings suggest that "trade" in Early Modern writing is a word that was primarily used in the context of institutional and commercial operations as well as national-level policy and international activities. However, there was also a lexically significant usage of trade denoting what a person does and skilled work. Using different methods across this project, I was able to delineate the coexistence of the two meanings within this corpus alongside their varying vocabularies and contexts.

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